

Full Derivation of Logistic Regression Gradient Descent

1 Model and Notation

The training set is

$$\{(x^{(i)}, y^{(i)}) \mid i = 1, 2, \dots, m\},$$

where

$$x^{(i)} = \begin{bmatrix} x_1^{(i)} \\ x_2^{(i)} \\ \vdots \\ x_n^{(i)} \end{bmatrix} \in \mathbb{R}^n, \quad y^{(i)} \in \{0, 1\}.$$

The parameters are

$$w = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \in \mathbb{R}^n, \quad b \in \mathbb{R}.$$

The logistic regression model is

$$z^{(i)} = w^\top x^{(i)} + b, \quad f_{w,b}(x^{(i)}) = g(z^{(i)}),$$

where $g(z)$ is the sigmoid function:

$$g(z) = \frac{1}{1 + e^{-z}}.$$

For brevity, we will often write

$$f^{(i)} \triangleq f_{w,b}(x^{(i)}), \quad z^{(i)} = w^\top x^{(i)} + b.$$

2 Cost Function

The loss (log loss) for a single training example $(x^{(i)}, y^{(i)})$ is defined as

$$\ell^{(i)}(w, b) = -[y^{(i)} \log f^{(i)} + (1 - y^{(i)}) \log(1 - f^{(i)})]. \quad (1)$$

The overall (average) cost function is

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m \ell^{(i)}(w, b) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log f^{(i)} + (1 - y^{(i)}) \log(1 - f^{(i)})]. \quad (2)$$

Our goal is to derive

$$\frac{\partial J}{\partial w_j} \quad \text{and} \quad \frac{\partial J}{\partial b},$$

and then write down the gradient descent update rules.

3 Sigmoid Derivative: $g'(z) = g(z)(1 - g(z))$

Step 1: Differentiate the definition

Start from the definition:

$$g(z) = \frac{1}{1 + e^{-z}}.$$

Rewrite it as a power to apply the chain rule easily:

$$g(z) = (1 + e^{-z})^{-1}.$$

Differentiate with respect to z :

$$g'(z) = -1 \cdot (1 + e^{-z})^{-2} \cdot \frac{d}{dz}(1 + e^{-z}).$$

Note that

$$\frac{d}{dz}(1 + e^{-z}) = 0 + \frac{d}{dz}e^{-z} = -e^{-z}.$$

Substituting, we obtain

$$\begin{aligned} g'(z) &= -(1 + e^{-z})^{-2} \cdot (-e^{-z}) \\ &= \frac{e^{-z}}{(1 + e^{-z})^2}. \end{aligned}$$

Step 2: Rewrite the derivative using $g(z)$

From the definition of $g(z)$,

$$g(z) = \frac{1}{1 + e^{-z}},$$

we get

$$1 - g(z) = 1 - \frac{1}{1 + e^{-z}} = \frac{1 + e^{-z} - 1}{1 + e^{-z}} = \frac{e^{-z}}{1 + e^{-z}}.$$

Therefore,

$$\begin{aligned} g(z)(1 - g(z)) &= \frac{1}{1 + e^{-z}} \cdot \frac{e^{-z}}{1 + e^{-z}} \\ &= \frac{e^{-z}}{(1 + e^{-z})^2}. \end{aligned}$$

This is exactly the same expression as we obtained for $g'(z)$, hence

$$\boxed{g'(z) = g(z)(1 - g(z))}. \quad (3)$$

4 Gradient of a Single Example w.r.t. w_j : $\partial \ell^{(i)} / \partial w_j$

Start from the single-example loss (1):

$$\ell^{(i)}(w, b) = -[y^{(i)} \log f^{(i)} + (1 - y^{(i)}) \log(1 - f^{(i)})].$$

Step 1: Differentiate $\ell^{(i)}$ w.r.t. $f^{(i)}$

Temporarily treat $f^{(i)}$ as a scalar variable f and write

$$\ell(f) = -[y \log f + (1 - y) \log(1 - f)].$$

Differentiate with respect to f :

$$\begin{aligned} \frac{d\ell}{df} &= - \left[y \cdot \frac{1}{f} + (1 - y) \cdot \frac{1}{1 - f} \cdot \frac{d}{df}(1 - f) \right] \\ &= - \left[\frac{y}{f} + (1 - y) \cdot \frac{1}{1 - f} \cdot (-1) \right] \\ &= - \left[\frac{y}{f} - \frac{1 - y}{1 - f} \right] \\ &= -\frac{y}{f} + \frac{1 - y}{1 - f}. \end{aligned}$$

Combine the two fractions:

$$\begin{aligned} \frac{d\ell}{df} &= \frac{1 - y}{1 - f} - \frac{y}{f} \\ &= \frac{(1 - y)f - y(1 - f)}{f(1 - f)} \\ &= \frac{f - yf - y + yf}{f(1 - f)} \\ &= \frac{f - y}{f(1 - f)}. \end{aligned}$$

Switch back to $f^{(i)}$ and $\ell^{(i)}$:

$$\frac{\partial \ell^{(i)}}{\partial f^{(i)}} = \frac{f^{(i)} - y^{(i)}}{f^{(i)}(1 - f^{(i)})}. \quad (4)$$

Step 2: $\partial f^{(i)} / \partial z^{(i)}$

Recall that

$$f^{(i)} = g(z^{(i)}).$$

Using (3), we have

$$\frac{\partial f^{(i)}}{\partial z^{(i)}} = g'(z^{(i)}) = g(z^{(i)}) (1 - g(z^{(i)})).$$

Since $g(z^{(i)}) = f^{(i)}$, this becomes

$$\frac{\partial f^{(i)}}{\partial z^{(i)}} = f^{(i)}(1 - f^{(i)}). \quad (5)$$

Step 3: $\partial z^{(i)} / \partial w_j$

We have

$$z^{(i)} = w^\top x^{(i)} + b = \sum_{k=1}^n w_k x_k^{(i)} + b.$$

When differentiating with respect to w_j , only the term $w_j x_j^{(i)}$ depends on w_j , hence

$$\frac{\partial z^{(i)}}{\partial w_j} = x_j^{(i)}. \quad (6)$$

Step 4: Chain rule to combine the three steps

By the chain rule,

$$\frac{\partial \ell^{(i)}}{\partial w_j} = \frac{\partial \ell^{(i)}}{\partial f^{(i)}} \cdot \frac{\partial f^{(i)}}{\partial z^{(i)}} \cdot \frac{\partial z^{(i)}}{\partial w_j}.$$

Substituting (4), (5), and (6) gives

$$\begin{aligned} \frac{\partial \ell^{(i)}}{\partial w_j} &= \frac{f^{(i)} - y^{(i)}}{f^{(i)}(1 - f^{(i)})} \cdot f^{(i)}(1 - f^{(i)}) \cdot x_j^{(i)} \\ &= (f^{(i)} - y^{(i)})x_j^{(i)}. \end{aligned}$$

Thus we obtain the gradient of the loss for a single example with respect to w_j :

$$\boxed{\frac{\partial \ell^{(i)}}{\partial w_j} = (f^{(i)} - y^{(i)})x_j^{(i)}}. \quad (7)$$

5 Gradient of the Overall Cost w.r.t. w_j : $\partial J / \partial w_j$

From the definition of $J(w, b)$ in (2):

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m \ell^{(i)}(w, b),$$

differentiating with respect to w_j gives

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m \frac{\partial \ell^{(i)}}{\partial w_j}.$$

Substitute (7):

$$\begin{aligned} \frac{\partial J}{\partial w_j} &= \frac{1}{m} \sum_{i=1}^m (f^{(i)} - y^{(i)})x_j^{(i)} \\ &= \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})x_j^{(i)}. \end{aligned}$$

Therefore,

$$\boxed{\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})x_j^{(i)}}. \quad (8)$$

6 Gradient w.r.t. the Bias b : $\partial J / \partial b$

The derivation for b is analogous to the derivation for w_j .

Step 1: Single-example gradient $\partial \ell^{(i)} / \partial b$

Again,

$$\ell^{(i)}(w, b) = \ell(f^{(i)}(z^{(i)}(w, b))),$$

so by the chain rule,

$$\frac{\partial \ell^{(i)}}{\partial b} = \frac{\partial \ell^{(i)}}{\partial f^{(i)}} \cdot \frac{\partial f^{(i)}}{\partial z^{(i)}} \cdot \frac{\partial z^{(i)}}{\partial b}.$$

The first two factors are still (4) and (5). We only need

$$z^{(i)} = w^\top x^{(i)} + b$$

differentiated with respect to b :

$$\frac{\partial z^{(i)}}{\partial b} = 1. \quad (9)$$

Substituting gives

$$\begin{aligned} \frac{\partial \ell^{(i)}}{\partial b} &= \frac{f^{(i)} - y^{(i)}}{f^{(i)}(1 - f^{(i)})} \cdot f^{(i)}(1 - f^{(i)}) \cdot 1 \\ &= f^{(i)} - y^{(i)}. \end{aligned}$$

Hence

$$\boxed{\frac{\partial \ell^{(i)}}{\partial b} = f^{(i)} - y^{(i)}}. \quad (10)$$

Step 2: Overall cost gradient

From

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m \ell^{(i)}(w, b),$$

we get

$$\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m \frac{\partial \ell^{(i)}}{\partial b}.$$

Substituting (10):

$$\begin{aligned} \frac{\partial J}{\partial b} &= \frac{1}{m} \sum_{i=1}^m (f^{(i)} - y^{(i)}) \\ &= \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}). \end{aligned}$$

Thus

$$\boxed{\frac{\partial J}{\partial b} = \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})}. \quad (11)$$

7 Gradient Descent Update Rules

The basic gradient descent update rule is

$$\theta := \theta - \alpha \frac{\partial J}{\partial \theta},$$

where α is the learning rate.

For w_j and b we have

$$w_j := w_j - \alpha \frac{\partial J}{\partial w_j}, \quad (12)$$

$$b := b - \alpha \frac{\partial J}{\partial b}. \quad (13)$$

Substituting (8) and (11):

$$w_j := w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x_j^{(i)} \right],$$
$$b := b - \alpha \left[\frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) \right].$$

In vector form, the updates can be written as

$$w := w - \alpha \cdot \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)},$$
$$b := b - \alpha \cdot \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}).$$

These are the full gradients and gradient descent update rules for logistic regression.